Penetrating Divisors: A Full Computation Ledger

Zeros and Poles as Diagnostic Instruments on $\mathbb{P}^1_{\mathbb{C}}$

A Computation-First Atlas

June 8, 2026

Every abstract object tied back to concrete local equations, valuations, bases, transition functions, and section evaluations: divisors, $L(D)$, $\mathcal{O}(D)$, Cartier data, line bundles, H^0, base points, and projective maps, computed in full.

A divisor is not merely a formal sum of points. A divisor is a diagnostic instrument. By placing poles where we wish to permit blow-up and zeros where we wish to force vanishing, we interrogate a curve and compel its geometry to answer. This document is a ledger: for each probe we do not assert what the targets are — we compute *them, step by step, from the divisor.*

1 The Situation and Its Targets

The situation. We analyze the complex projective line $X = \mathbb{P}^1_{\mathbb{C}}$, covered by the two standard affine charts

$$U_0 = \mathbb{P}^1 \setminus \{\infty\} \text{ (coordinate } z), \qquad U_1 = \mathbb{P}^1 \setminus \{0\} \text{ (coordinate } w = \tfrac{1}{z}),$$

with overlap $U_0 \cap U_1 = \mathbb{C}^\times$. The local coordinate at a finite point a is $t_a = z - a$; at ∞ it is $t_\infty = w$. For a rational function $f = N(z)/M(z)$ in lowest terms,

$$\operatorname{ord}_a(f) = \operatorname{ord}_a N - \operatorname{ord}_a M \quad (a \text{ finite}), \qquad \operatorname{ord}_\infty(f) = \deg M - \deg N.$$

We probe X with divisors supported in $\{0, \infty\}$, so each support point is the center of exactly one chart and the two-chart cover is automatically adapted to the support (see the remark below).

Remark 1 (Stated assumptions)**.** The situation slot was delivered empty. To honour the requirement that every quantity be *computed*, we fix $X = \mathbb{P}^1_{\mathbb{C}}$ — the setting where $L(D)$, $\mathcal{O}(D)$, Cartier data, transition functions, and projective maps all admit closed-form rational answers. We further place all supports in $\{0, \infty\}$; then the support-adapted cover prescribed in general (an open $U_0 = X \setminus \operatorname{Supp} D$ together with disks U_p around each p, with local equation $f_p = t_p^{n_p}$) coincides with the two affine charts: the chart U_0 carries the support at 0 via $f_0 = z^{n_0} = t_0^{n_0}$, and the chart U_1 carries the support at ∞ via $f_1 = w^{n_\infty} = t_\infty^{n_\infty}$. This keeps every transition function a single monomial.

The targets. For each divisor we expose, in order:

1. the divisor D itself, read as valuation inequalities;
2. the Riemann–Roch space $L(D)$ with an explicit basis;
3. the invertible sheaf $\mathcal{O}(D)$;
4. the Cartier local equations f_α;
5. the transition functions $g_{\alpha\beta} = f_\beta / f_\alpha$;
6. the associated holomorphic line bundle;

7. the global sections $H^0(X, \mathcal{O}(D))$ and their local representatives;
8. the base points of the linear system $|D|$;
9. the projective map Φ_D;
10. separation of points and tangent directions.

Each probe receives a *full calculation ledger* computing all ten targets explicitly. Nothing is asserted that is not first computed.

2 Selection of Penetrating Divisors

The probes are ordered by degree, sweeping from below the floor of visibility up to a genuine embedding, and ending with the one experiment a *complete* linear system on $\mathbb{P}^1$ can never perform.

Probe A: the void. $D_1 = -[0]$ (degree -1). What does a *negative* divisor do to global sections? It forbids everything. Tests whether H^0 can be empty.

Probe B: the collapse. $D_2 = [\infty]-[0]$ (degree 0, nonzero). Exposes linear equivalence $\mathcal{O}(D_2) \cong \mathcal{O}$ and the gap between a *raw value* and a *trivialized section value*.

Probe C: the minimal embedding. $D_3 = [\infty]$ (degree 1). Realizes $\mathcal{O}(1)$ as very ample and embeds $\mathbb{P}^1$ as a line.

Probe D: the conic. $D_4 = 2[\infty]$ (degree 2). Embeds $\mathbb{P}^1$ as a smooth conic in $\mathbb{P}^2$ (the rational normal curve); tests separation of points *and* tangents.

Probe E: the incomplete cut. subsystems $V \subsetneq H^0(\mathbb{P}^1, \mathcal{O}(2[\infty]))$. Manufactures the two failures a complete system on $\mathbb{P}^1$ cannot show: a *base point* and a *failure to separate points*.

3 The Calculation Protocol and Common Machinery

Each ledger executes Targets 1–10 of Section 1. Two computations recur in every probe, so we record them once.

Uniform Cartier data and transition function. For $D = a[0] + b[\infty]$ with $d := \deg D = a + b$, the support-adapted equations are

$$f_0 = z^{\,a} \text{ on } U_0, \qquad f_1 = w^{\,b} \text{ on } U_1.$$

On the overlap $\mathbb{C}^\times$, using $w = 1/z$,

$$\frac{f_0}{f_1} = \frac{z^a}{w^b} = z^a\, z^b = z^{\,a+b} = z^{\,d}, \qquad g_{01} = \frac{f_1}{f_0} = z^{-d} = w^{\,d} \in \mathcal{O}^\times(\mathbb{C}^\times),$$

because the only zero/pole of z^d sits at $z = 0$ or $z = \infty$, both deleted from $\mathbb{C}^\times$. Hence $\mathcal{O}(D) \cong \mathcal{O}(d)$.

Riemann–Roch on $\mathbb{P}^1$. Here $g = 0$ and the canonical divisor is $K = -2[\infty]$ ($\deg K = 2g - 2 = -2$), so

$$\ell(D) - \ell(K - D) = \deg D + 1 - g = \deg D + 1.$$

If $\deg D \geq -1$ then $\deg(K - D) = -2 - \deg D \leq -1 < 0$, forcing $\ell(K - D) = 0$ and

$$\boxed{\ell(D) = \deg D + 1 \quad (\deg D \geq -1)}, \qquad \ell(D) = 0 \quad (\deg D < 0).$$

Every ledger cross-checks this count against the basis it derives.

4 Full Calculation Ledgers

Full Calculation Ledger for $D_1 = -[0]$



$D_1 = -[0]$: one forced zero at $z = 0$; no pole permitted anywhere; $\deg D_1 = -1$.

TARGET 1. **The divisor itself**

$D_1 = (-1)\,[0] = \sum_p n_p[p]$ with $n_0 = -1$ and $n_p = 0$ otherwise. The defining inequality $\mathrm{div}(f) + D_1 \geq 0$ is read pointwise as $\mathrm{ord}_p(f) + n_p \geq 0$:

point p	coeff. n_p	condition $\mathrm{ord}_p(f) + n_p \geq 0$	meaning
0	-1	$\mathrm{ord}_0(f) \geq 1$	f must vanish to order ≥ 1 at 0
∞	0	$\mathrm{ord}_\infty(f) \geq 0$	f holomorphic at ∞
other p	0	$\mathrm{ord}_p(f) \geq 0$	f holomorphic at p

$\deg D_1 = \sum_p n_p = -1$.

TARGET 2. **The Riemann–Roch space $L(D_1)$**

By the table, $f \in L(D_1)$ is holomorphic at every point of $\mathbb{P}^1$ (no poles anywhere) and vanishes at 0. A function holomorphic on all of the compact $\mathbb{P}^1$ is constant; the only constant with $\mathrm{ord}_0 \geq 1$ is 0. Hence

$$L(D_1) = \{f \in \mathbb{C}(z) : \mathrm{ord}_0(f) \geq 1,\ \mathrm{ord}_p(f) \geq 0\ \forall p\} = \{0\}, \qquad \ell(D_1) = 0.$$

Riemann–Roch check. $K - D_1 = -2[\infty] + [0]$ has degree $-1 < 0$, so $\ell(K - D_1) = 0$ and $\ell(D_1) = \deg D_1 + 1 = 0$. Exact. There is no basis to tabulate.

TARGET 3. **The sheaf $\mathcal{O}(D_1)$**

$\mathcal{O}(D_1)(U) = \{f \in \mathcal{M}(U) : \mathrm{div}(f) + D_1|_U \geq 0\}$. Locally:

- near 0 (negative coefficient): sections are forced to vanish, $\mathcal{O}(D_1)(U_0^{\mathrm{sm}}) = \{f : \mathrm{ord}_0(f) \geq 1\} = z \cdot \mathcal{O}_0$, the maximal ideal $\mathfrak{m}_0$;
- away from 0 (e.g. near ∞ or any $p \neq 0$): no condition, $\mathcal{O}(D_1)(U) = \mathcal{O}(U)$.

Taking $U = X$ reproduces Target 2: $H^0(X, \mathcal{O}(D_1)) = \mathcal{O}(D_1)(X) = L(D_1) = 0$, since a global section is exactly a global meromorphic function meeting every valuation inequality of D_1.

TARGET 4. **Cartier data**

Here $a = n_0 = -1,\ b = n_\infty = 0$, so

$$f_0 = z^{-1} \text{ on } U_0, \qquad f_1 = w^0 = 1 \text{ on } U_1.$$

Check $\mathrm{div}(f_0)|_{U_0}$: z^{-1} has a single pole at $z = 0$ and no other zero/pole in U_0, so $\mathrm{div}(z^{-1})|_{U_0} = -[0] = D_1|_{U_0}$. Check $\mathrm{div}(f_1)|_{U_1}$: $0 \notin U_1$, so $D_1|_{U_1} = 0 = \mathrm{div}(1)$. On the overlap,

$$\frac{f_0}{f_1} = z^{-1},$$

whose pole ($z = 0$) and zero ($z = \infty$) both lie outside $\mathbb{C}^\times$; hence $f_0/f_1 \in \mathcal{O}^\times(\mathbb{C}^\times)$.

TARGET 5. **Transition functions**

Local generators $e_\alpha = f_\alpha^{-1}$:

$$e_0 = f_0^{-1} = z, \qquad e_1 = f_1^{-1} = 1.$$

From $e_0 = (f_1/f_0)e_1$ the transition function is

$$g_{01} = \frac{f_1}{f_0} = \frac{1}{z^{-1}} = z = w^{-1} \quad (= z^{-d} \text{ with } d = -1).$$

With only two charts the cocycle conditions are $g_{00} = g_{11} = 1$ and $g_{10} = f_0/f_1 = z^{-1} = g_{01}^{-1}$; triple overlaps are empty, so the cocycle identity $g_{\alpha\beta}g_{\beta\gamma} = g_{\alpha\gamma}$ holds vacuously.

TARGET 6. **Invertible sheaf and line bundle**

$\mathcal{O}(D_1)|_{U_0} = \mathcal{O}_{U_0}\, e_0$ and $\mathcal{O}(D_1)|_{U_1} = \mathcal{O}_{U_1}\, e_1$ are free of rank 1. The line bundle $L_1 \to \mathbb{P}^1$ is $U_0 \times \mathbb{C}$ glued to $U_1 \times \mathbb{C}$ by $v_0 = g_{01}v_1 = z\, v_1$; this is $\mathcal{O}(-1)$, the tautological bundle. *Linear equivalence:* with $h = 1/z$, $\operatorname{div}(h) = \operatorname{div}(1/z) = [\infty] - [0]$, so $D_1 - (-[\infty]) = -[0] + [\infty] = \operatorname{div}(1/z)$, giving $D_1 \sim -[\infty]$ and again $\mathcal{O}(D_1) \cong \mathcal{O}(-1)$.

TARGET 7. **Global sections and local representatives**

$H^0 = 0$, so there is no nonzero section to trivialize. The mechanism is visible in the generators: $e_0 = z$ is holomorphic on U_0, but read on U_1 it is $e_0 = z = w^{-1}$, which has a pole at $w = 0$ ($z = \infty$). No nonzero constant multiple of e_0 extends across U_1, which is exactly why the global section space is empty.

TARGET 8. **Base points**

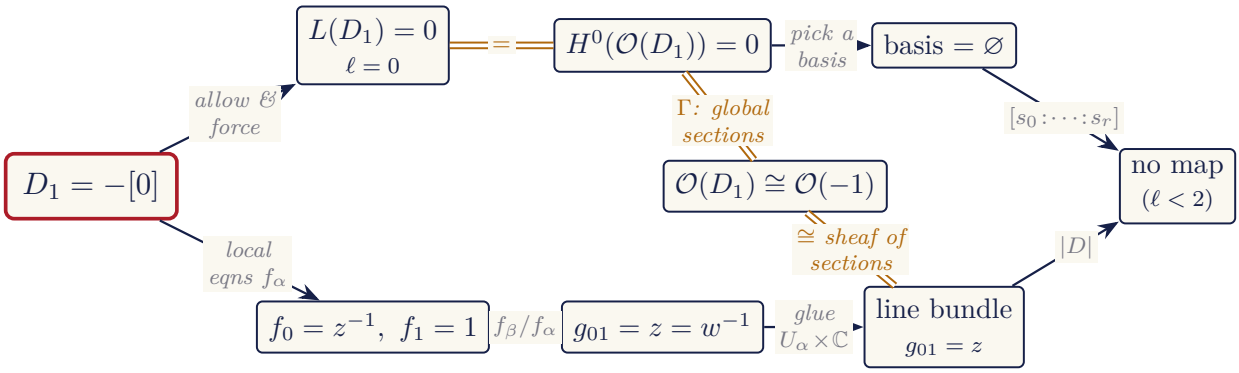
The complete linear system $|D_1| = \{E \geq 0 : E \sim D_1\}$ is *empty*: any $E \sim D_1$ has $\deg E = \deg D_1 = -1 < 0$, impossible for an effective divisor. Equivalently $\dim H^0 = 0$. There is no section, hence no well-defined base-point test; conventionally every point is a (trivial) base point of the empty system.

TARGET 9. **Projective map**

Since $\dim H^0(X, \mathcal{O}(D_1)) = 0 < 2$, no map to any $\mathbb{P}^r$ is defined.

TARGET 10. **Separation of points and tangent directions**

Not applicable: there is no morphism. The probe's content is precisely the *absence* of all downstream structure once $\deg D < 0$.



Probe tests: whether H^0 can vanish.
It reveals: $\mathcal{O}(-1)$ has no global sections; a negative divisor sees nothing.
Verdict: The void — no sections, no map.

Full Calculation Ledger for $D_2 = [\infty] - [0]$



$D_2 = [\infty] - [0]$: forced zero at 0, simple pole allowed at ∞; $\deg D_2 = 0$.

TARGET 1. **The divisor itself**

$D_2 = (-1)[0] + (+1)[\infty]$, so $n_0 = -1,\ n_\infty = +1$:

point p	coeff. n_p	condition $\mathrm{ord}_p(f)+n_p\geq 0$	meaning
0	-1	$\mathrm{ord}_0(f)\geq 1$	f vanishes to order ≥ 1 at 0
∞	$+1$	$\mathrm{ord}_\infty(f)\geq -1$	f may have a pole of order ≤ 1 at ∞
other p	0	$\mathrm{ord}_p(f)\geq 0$	f holomorphic at p

$\deg D_2 = -1+1 = 0$.

Target 2. The Riemann–Roch space $L(D_2)$

f has no finite pole (the only finite support point, 0, *forces a zero*), so f is a polynomial; the bound $\mathrm{ord}_\infty(f)\geq -1$ means $\deg f\leq 1$; and $\mathrm{ord}_0(f)\geq 1$ forces $f(0)=0$. A degree-≤ 1 polynomial vanishing at 0 is $f = c\,z$. Therefore

$$L(D_2)=\mathbb{C}\cdot z, \qquad \ell(D_2)=1=\deg D_2+1\ \checkmark.$$

Basis $s=z$:

basis s	ord_0	ord_∞	$\mathrm{div}(s)+D_2$	≥ 0?
z	1	-1	$([0]-[\infty])+([\infty]-[0])=0$	yes

Target 3. The sheaf $\mathcal{O}(D_2)$

Local sections: near 0, $\{f : \mathrm{ord}_0(f)\geq 1\}$ (vanishing); near ∞, $\{f : \mathrm{ord}_\infty(f)\geq -1\}$ (pole ≤ 1); near any other p, holomorphic. Globally $H^0(X,\mathcal{O}(D_2)) = L(D_2) = \mathbb{C}z$.

Target 4. Cartier data

$a=-1,\ b=+1$:

$$f_0 = z^{-1} \text{ on } U_0, \qquad f_1 = w^1 = w \text{ on } U_1.$$

$\mathrm{div}(z^{-1})|_{U_0} = -[0] = D_2|_{U_0}$ (the $+[\infty]$ is outside U_0); $\mathrm{div}(w)|_{U_1} = +[\infty] = D_2|_{U_1}$ (the $-[0]$ is outside U_1). Overlap:

$$\frac{f_0}{f_1}=\frac{z^{-1}}{w}=\frac{z^{-1}}{z^{-1}}=1\in\mathcal{O}^\times(\mathbb{C}^\times),$$

a nowhere-zero constant.

Target 5. Transition functions

$e_0 = f_0^{-1} = z,\ e_1 = f_1^{-1} = w^{-1} = z$, and

$$g_{01}=\frac{f_1}{f_0}=\frac{w}{z^{-1}}=w\,z=1 \quad (=z^{-d},\ d=0).$$

The transition function is the constant 1: the bundle is trivial. Cocycle vacuous on (empty) triple overlaps.

Target 6. Invertible sheaf and line bundle

$g_{01}=1$ glues $U_0\times\mathbb{C}$ and $U_1\times\mathbb{C}$ into the *trivial* bundle $\mathbb{P}^1\times\mathbb{C}$, so $\mathcal{O}(D_2)\cong\mathcal{O}$. *Linear equivalence:* D_2 is principal, $D_2=\mathrm{div}(1/z)$ since $\mathrm{div}(1/z)=[\infty]-[0]$; thus $D_2\sim 0$ and $\mathcal{O}(D_2)\cong\mathcal{O}(0)=\mathcal{O}$, consistent with $\deg D_2=0$.

Target 7. Global sections and local representatives

For the basis $s=z$, $a_\alpha = f_\alpha s$:

section s	$a_0 = f_0 s$ on U_0	$a_1 = f_1 s$ on U_1 (in w)	holomorphic?
z	$z^{-1}\cdot z = 1$	$w\cdot z = w\cdot\frac{1}{w}=1$	yes (both)

Warning. *The raw meromorphic value of a section is not the same thing as the value of that section in a local trivialization.* The raw function z vanishes at 0 ($z(0) = 0$) and blows up at ∞; yet its *section* has trivialized values $a_0 = 1$ at 0 and $a_1 = 1$ at ∞ — *nowhere zero.* This is the nowhere-vanishing section of the trivial bundle.

Target 8. Base points

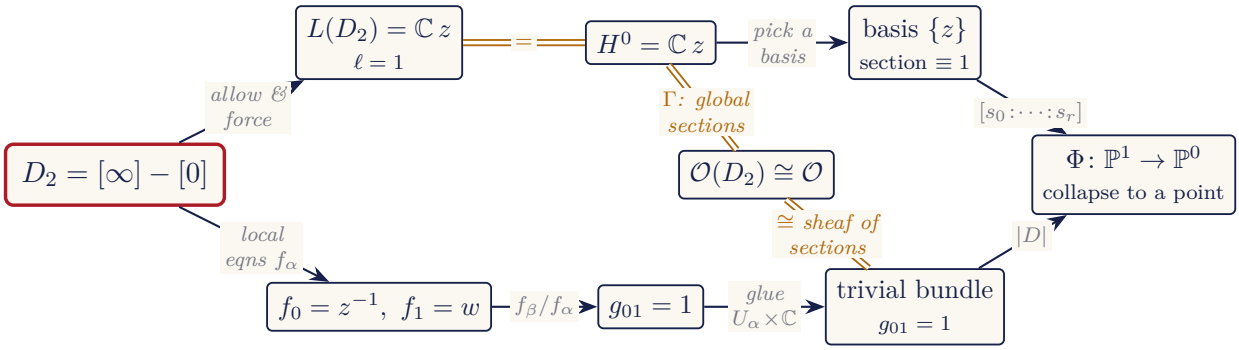
The single section $s = z$ has trivialized values $a_0 = a_1 = 1 \neq 0$ everywhere, so $|D_2|$ has *no base points.* (Deciding this from the raw value $z(0) = 0$ would be wrong — hence the warning above.)

Target 9. Projective map

$\dim H^0 = 1$, so $\Phi_{D_2} : \mathbb{P}^1 \to \mathbb{P}^0 = \{\text{pt}\}$ is the constant map: every point goes to the single point of $\mathbb{P}^0$.

Target 10. Separation of points and tangent directions

A constant map separates nothing: it collapses all of $\mathbb{P}^1$ to a point. It is not an embedding (its image is 0-dimensional). The degree-0 probe sees the trivial bundle but no geometry.



Probe tests: raw value vs. trivialized value.

It reveals: a trivial bundle whose generating section is the meromorphic function z, nowhere zero; the map collapses to a point.

Verdict: Collapse — a section exists, the map is constant.

Full Calculation Ledger for $D_3 = [\infty]$



$D_3 = [\infty]$: simple pole allowed at ∞, no forced zeros; $\deg D_3 = 1$.

Target 1. The divisor itself

$D_3 = (+1)[\infty]$, $n_\infty = +1$:

point p	coeff. n_p	condition $\text{ord}_p(f) + n_p \geq 0$	meaning
∞	$+1$	$\text{ord}_\infty(f) \geq -1$	pole of order ≤ 1 allowed at ∞
other p	0	$\text{ord}_p(f) \geq 0$	f holomorphic at p

$\deg D_3 = 1$.

Target 2. The Riemann–Roch space $L(D_3)$

No finite poles $\Rightarrow$ f a polynomial; $\text{ord}_\infty(f) \geq -1 \Rightarrow \deg f \leq 1$. Hence $f = c_0 + c_1 z$ and

$$L(D_3) = \text{span}_{\mathbb{C}}\{1,\ z\}, \qquad \ell(D_3) = 2 = \deg D_3 + 1 \ \checkmark.$$

basis s	ord_0	ord_∞	$\text{div}(s) + D_3$	≥ 0?
1	0	0	$0 + [\infty] = [\infty]$	yes
z	1	-1	$([0] - [\infty]) + [\infty] = [0]$	yes

TARGET 3. **The sheaf $\mathcal{O}(D_3)$**

Near ∞, sections may have a simple pole; everywhere else holomorphic. $H^0(X, \mathcal{O}(D_3)) = L(D_3) = \langle 1, z \rangle$.

TARGET 4. **Cartier data**

$a = 0,\ b = 1$:

$$f_0 = z^0 = 1 \text{ on } U_0, \qquad f_1 = w \text{ on } U_1.$$

$\operatorname{div}(1)|_{U_0} = 0 = D_3|_{U_0}$ ($\infty \notin U_0$); $\operatorname{div}(w)|_{U_1} = [\infty] = D_3|_{U_1}$. Overlap: $f_0/f_1 = 1/w = z$, with zero at 0 and pole at ∞ both outside $\mathbb{C}^\times$, so $f_0/f_1 \in \mathcal{O}^\times(\mathbb{C}^\times)$.

TARGET 5. **Transition functions**

$e_0 = 1,\ e_1 = w^{-1} = z$, and

$$g_{01} = \frac{f_1}{f_0} = w = z^{-1} \quad (= z^{-d},\ d = 1).$$

Cocycle: $g_{10} = z = g_{01}^{-1}$, triple overlaps empty (vacuous).

TARGET 6. **Invertible sheaf and line bundle**

Gluing $U_0 \times \mathbb{C}$, $U_1 \times \mathbb{C}$ by $g_{01} = w$ produces $\mathcal{O}(1)$, the hyperplane bundle. *Linear equivalence:* $[\infty] - [0] = \operatorname{div}(1/z)$ gives $[\infty] \sim [0]$, and more generally $[\infty] \sim [p]$ for every p via $\operatorname{div}(\frac{z-p}{1}) = [p] - [\infty]$; all are $\mathcal{O}(1)$.

TARGET 7. **Global sections and local representatives**

$a_\alpha = f_\alpha s$ for $s \in \{1, z\}$:

section s	$a_0 = f_0 s$ on U_0	$a_1 = f_1 s$ on U_1 (in w)	holomorphic?
1	1	$w \cdot 1 = w$	yes
z	z	$w \cdot z = w \cdot \frac{1}{w} = 1$	yes

Warning. *The raw meromorphic value of a section is not the same thing as the value of that section in a local trivialization.* Raw z is "∞" at ∞, but its trivialized value there is $a_1 = 1 \neq 0$.

TARGET 8. **Base points**

On U_0 (trivialization $f_0 = 1$) the sections read $(1, z)$; the entry 1 never vanishes, so no finite base point. On U_1 ($f_1 = w$) they read $(w, 1)$; at ∞ ($w = 0$) this is $(0, 1)$, and the second entry is nonzero. No base points: $|D_3|$ is base-point-free.

TARGET 9. **Projective map**

$\dim H^0 = 2$, so $\Phi_{D_3} \colon \mathbb{P}^1 \to \mathbb{P}^1$. On U_0,

$$\Phi_{D_3}(z) = [\, s_0(z) : s_1(z) \,] = [\, 1 : z \,].$$

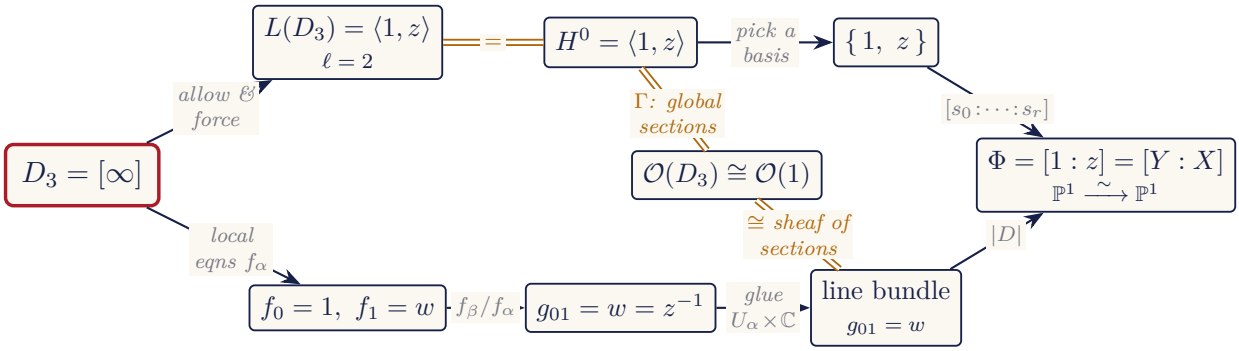
At ∞, use the U_1 representatives: $[a_1(s_0) : a_1(s_1)] = [w : 1]\big|_{w=0} = [0 : 1]$. In homogeneous coordinates $[X : Y]$ with $z = X/Y$ and $\infty = [1 : 0]$,

$$\Phi_{D_3}([X : Y]) = [\, Y : X \,],$$

i.e. $[1 : z] = [Y : X]$ after clearing Y. The component polynomials $F_0 = Y, F_1 = X$ are linear with no common zero, confirming base-point-freeness and $\deg \Phi = 1$.

TARGET 10. **Separation of points and tangent directions**

The matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has determinant $-1 \neq 0$, so $\Phi_{D_3} \in \mathrm{PGL}_2(\mathbb{C})$ is an *automorphism* (the coordinate swap $z \leftrightarrow 1/\cdot$). It is bijective (separates points) and an isomorphism on tangent spaces (separates tangents); hence an embedding — indeed an isomorphism $\mathbb{P}^1 \xrightarrow{\sim} \mathbb{P}^1$. This is the minimal very ample bundle $\mathcal{O}(1)$.



Probe tests: base-point-freeness and very ampleness.
It reveals: $\mathbb{P}^1$ embeds as a line via $\mathcal{O}(1)$; the map is the automorphism $[X : Y] \mapsto [Y : X]$.
Verdict: Embedding (isomorphism).

Full Calculation Ledger for $D_4 = 2[\infty]$



$D_4 = 2[\infty]$: pole of order ≤ 2 allowed at ∞, no forced zeros; $\deg D_4 = 2$.

Target 1. The divisor itself

$D_4 = (+2)[\infty]$, $n_\infty = +2$:

point p	coeff. n_p	condition $\mathrm{ord}_p(f) + n_p \geq 0$	meaning
∞	$+2$	$\mathrm{ord}_\infty(f) \geq -2$	pole of order ≤ 2 allowed at ∞
other p	0	$\mathrm{ord}_p(f) \geq 0$	f holomorphic at p

$\deg D_4 = 2$.

Target 2. The Riemann–Roch space $L(D_4)$

No finite poles $\Rightarrow$ polynomial; $\mathrm{ord}_\infty(f) \geq -2 \Rightarrow \deg f \leq 2$. Hence $f = c_0 + c_1 z + c_2 z^2$ and

$$L(D_4) = \mathrm{span}_{\mathbb{C}}\{1,\ z,\ z^2\}, \qquad \ell(D_4) = 3 = \deg D_4 + 1\ \checkmark.$$

basis s	ord_0	ord_∞	$\mathrm{div}(s) + D_4$	≥ 0?
1	0	0	$2[\infty]$	yes
z	1	-1	$([0] - [\infty]) + 2[\infty] = [0] + [\infty]$	yes
z^2	2	-2	$(2[0] - 2[\infty]) + 2[\infty] = 2[0]$	yes

Target 3. The sheaf $\mathcal{O}(D_4)$

Near ∞, poles of order ≤ 2 allowed; elsewhere holomorphic. $H^0(X, \mathcal{O}(D_4)) = L(D_4) = \langle 1, z, z^2\rangle$.

Target 4. Cartier data

$a = 0,\ b = 2$:

$$f_0 = 1 \text{ on } U_0, \qquad f_1 = w^2 \text{ on } U_1.$$

$\mathrm{div}(1)|_{U_0} = 0$; $\mathrm{div}(w^2)|_{U_1} = 2[\infty] = D_4|_{U_1}$. Overlap: $f_0/f_1 = 1/w^2 = z^2$, zero at 0 and pole at ∞ both outside $\mathbb{C}^\times$, so $\in \mathcal{O}^\times(\mathbb{C}^\times)$.

Target 5. Transition functions

$e_0 = 1,\ e_1 = w^{-2} = z^2$, and

$$g_{01} = \frac{f_1}{f_0} = w^2 = z^{-2} \quad (= z^{-d},\ d = 2).$$

Cocycle: $g_{10} = z^2 = g_{01}^{-1}$, triple overlaps empty (vacuous).

Target 6. Invertible sheaf and line bundle

Gluing by $g_{01} = w^2$ yields $\mathcal{O}(2)$. *Linear equivalence:* $2[\infty] - ([0] + [\infty]) = [\infty] - [0] = \operatorname{div}(1/z)$, so $2[\infty] \sim [0] + [\infty] \sim 2[0]$ — all degree-2 divisors, all $\mathcal{O}(2)$.

Target 7. Global sections and local representatives

$a_\alpha = f_\alpha s$ for $s \in \{1, z, z^2\}$:

section s	$a_0 = f_0 s$ on U_0	$a_1 = f_1 s$ on U_1 (in w)	holomorphic?
1	1	$w^2 \cdot 1 = w^2$	yes
z	z	$w^2 \cdot z = w$	yes
z^2	z^2	$w^2 \cdot z^2 = 1$	yes

Warning. *The raw meromorphic value of a section is not the same thing as the value of that section in a local trivialization.* At ∞ the raw values of $1, z, z^2$ are $1, \infty, \infty$, but the trivialized representatives are $w^2, w, 1$, i.e. $0, 0, 1$: only z^2 survives, and it does *not* vanish.

Target 8. Base points

On U_0, sections $(1, z, z^2)$: the entry 1 never vanishes $\Rightarrow$ no finite base point. On U_1, representatives $(w^2, w, 1)$: at ∞ $(w = 0)$ this is $(0, 0, 1)$, third entry nonzero $\Rightarrow$ ∞ not a base point. $|D_4|$ is base-point-free.

Target 9. Projective map

$\dim H^0 = 3$, so $\Phi_{D_4} : \mathbb{P}^1 \to \mathbb{P}^2$. On U_0,

$$\Phi_{D_4}(z) = [\,1 : z : z^2\,].$$

At ∞, U_1 representatives give $[w^2 : w : 1]\big|_{w=0} = [0 : 0 : 1]$. In homogeneous coordinates $(z = X/Y)$,

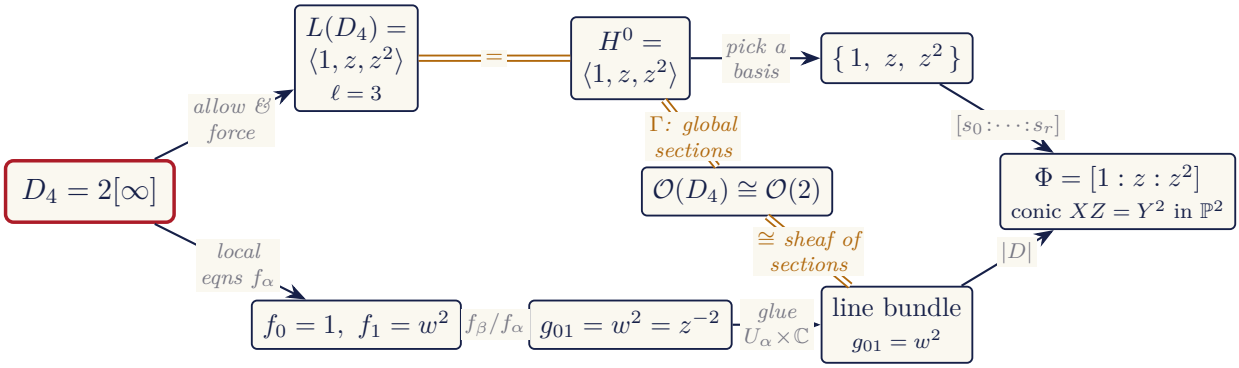
$$\Phi_{D_4}([X : Y]) = [\,Y^2 : XY : X^2\,],$$

the 2nd Veronese map ν_2, of degree 2. Its image is the conic

$$C = \{[A : B : C] \in \mathbb{P}^2 : AC = B^2\}, \qquad (A, B, C) = (Y^2, XY, X^2), \quad AC = X^2Y^2 = B^2.$$

Target 10. Separation of points and tangent directions

Points: from $[1 : z : z^2]$ the ratio (2nd)/(1st)$= z$ recovers z uniquely, and $\infty \mapsto [0 : 0 : 1]$ is hit by no finite z; so Φ_{D_4} is injective. *Tangents:* on U_0 the affine map $z \mapsto (z, z^2)$ has derivative $(1, 2z) \neq (0, 0)$; near ∞, in the chart $C \neq 0$, it is $w \mapsto (w^2, w)$ with derivative $(2w, 1) \neq (0, 0)$. Both are immersions, so tangent directions are separated. Hence Φ_{D_4} is a closed embedding; $\mathbb{P}^1 \cong C$, a smooth conic — the rational normal curve of degree 2 (Veronese).



Probe tests: separation of points and tangent directions.
It reveals: $\mathcal{O}(2)$ is very ample; the rational normal curve embeds $\mathbb{P}^1$ as a smooth conic $[X : Y] \mapsto [Y^2 : XY : X^2]$.
Verdict: Embedding (conic).

Full Calculation Ledger for the incomplete cuts of $|2[\infty]|$



Same bundle $\mathcal{O}(2)$ as D_4; we keep only a 2-dimensional subspace $V \subsetneq H^0$.

TARGET 1–6. **Bundle data: identical to D_4**

A linear system is a pair (line bundle, subspace of sections). The underlying divisor class is still $2[\infty]$, so Targets 1–6 are *unchanged* from the D_4 ledger:

$$\deg = 2, \quad f_0 = 1,\ f_1 = w^2, \quad g_{01} = w^2, \quad \mathcal{O} \cong \mathcal{O}(2), \quad H^0 = \langle 1, z, z^2 \rangle, \quad \ell = 3.$$

The novelty is at Targets 7–10: we replace the complete H^0 by a proper subspace and watch the consequences. On $\mathbb{P}^1$ every complete $|D|$ with $\deg \geq 0$ is base-point-free, so this is the only way to exhibit a base point or a non-injective map.

Cut E1: $V_{\text{bp}} = \text{span}_{\mathbb{C}}\{z,\ z^2\} = z \cdot \langle 1, z \rangle$.

TARGET 7. **Local representatives**

On U_0 ($f_0 = 1$) the representatives are literally $(z,\ z^2)$; on U_1 ($f_1 = w^2$) they are $(w^2 z,\ w^2 z^2) = (w,\ 1)$.

TARGET 8. **Base points**

On U_0, evaluate the trivialized vector at $z = 0$: $(z, z^2)\big|_0 = (0, 0)$. *Both* representatives vanish, so $z = 0$ is a **base point**. (This is a genuine trivialized computation, not a raw-value artifact: on U_0, $f_0 = 1$, so raw and trivialized values coincide here.) The base locus is the common zero divisor $\gcd(z, z^2) = z$, i.e. the fixed part is $[0]$; the moving part is $|2[\infty] - [0]| \cong |\mathcal{O}(1)|$.

TARGET 9. **Projective map**

Away from the base point, $[z : z^2] = [1 : z]$, so V_{bp} defines only a *rational* map

$$\mathbb{P}^1 \dashrightarrow \mathbb{P}^1, \qquad z \mapsto [z : z^2] = [1 : z] \quad (z \neq 0),$$

undefined at $z = 0$ (homogeneously $[X : Y] \mapsto [XY : X^2] = [Y : X]$, indeterminate at $[0 : 1]$). After blowing down the fixed part $[0]$ the residual map is the degree-1 map of Probe C.

TARGET 10. **Separation**

Not everywhere defined: it fails condition (1) outright because of the base point. The deleted section was $s_0 = 1$ — the unique section nonzero at 0 — so removing it created the base point exactly there.

Cut E2: $V_{\text{sep}} = \text{span}_{\mathbb{C}}\{1,\ z^2\}$.

TARGET 7. **Local representatives**

On U_0: $(1,\ z^2)$. On U_1: $(w^2 \cdot 1,\ w^2 \cdot z^2) = (w^2,\ 1)$.

TARGET 8. **Base points**

On U_0 the entry 1 never vanishes; on U_1 at ∞ the vector is $(0, 1)$, second entry nonzero. No base points: V_{sep} is base-point-free, so it defines an honest *morphism*.

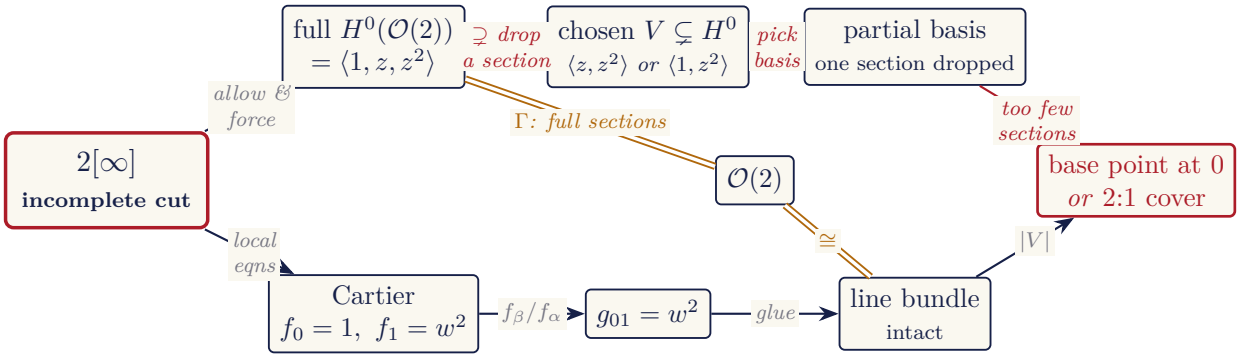
TARGET 9. **Projective map**

$$\psi(z) = [\,1 : z^2\,] \colon \mathbb{P}^1 \to \mathbb{P}^1, \qquad \psi([X : Y]) = [\,Y^2 : X^2\,],$$

of degree 2.

TARGET 10. **Separation**

Points: $\psi(z) = \psi(-z)$ since $[1 : z^2] = [1 : (-z)^2]$, so ψ is 2:1 and does *not* separate the points z and $-z$. *Tangents:* on U_0, $z \mapsto z^2$ has derivative $2z$, vanishing at $z = 0$; so ψ fails to separate tangents at 0 (and likewise at ∞). Thus ψ is a finite degree-2 morphism (the squaring map, branched at 0 and ∞) — a finite cover, *not* an embedding, even though it is base-point-free.



Map E: the bottom rail and the sheaf $\mathcal{O}(2)$ stay intact (amber rungs hold). The whole failure is the single red step $H^0 \supsetneq V$ — dropping a section — which propagates into a base point (V_{bp}) or a 2:1 cover (V_{sep}).

Probe tests: whether base-point-freeness suffices for an embedding.
It reveals: it does not: $\langle z, z^2 \rangle$ has a base point at 0; $\langle 1, z^2 \rangle$ is base-point-free yet a branched double cover.
Verdict: Not an embedding — completeness was doing real work.

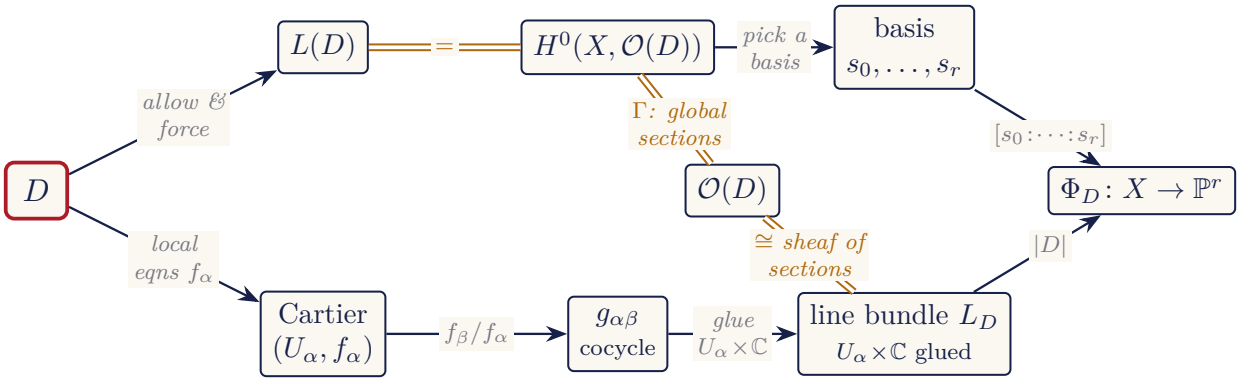
5 Comparative Table

The five instruments side by side. Rows are the targets; columns are the probes.

	A: $-[0]$	**B:** $[\infty]-[0]$	**C:** $[\infty]$	**D:** $2[\infty]$	**E:** sub-$\lvert 2[\infty] \rvert$
Degree d	-1	0	1	2	2 (bundle)
Allowed poles	none	≤ 1 at ∞	≤ 1 at ∞	≤ 2 at ∞	≤ 2 at ∞
Forced zeros	≥ 1 at 0	≥ 1 at 0	none	none	E1: ≥ 1 at 0
$\dim L = \ell$	0	1	2	3	2 (subspace)
Basis	$\varnothing$	z	$1,\ z$	$1,\ z,\ z^2$	$z, z^2\ /\ 1, z^2$
Cartier $f_0; f_1$	$z^{-1};\ 1$	$z^{-1};\ w$	$1;\ w$	$1;\ w^2$	$1;\ w^2$
Transition g_{01}	z	1	w	w^2	w^2
$\mathcal{O}(D)$	$\mathcal{O}(-1)$	$\mathcal{O}$	$\mathcal{O}(1)$	$\mathcal{O}(2)$	$\mathcal{O}(2)$
Base points?	— (empty)	none	none	none	$\{0\}$ / none
Projective map	none	to a point	$[X{:}Y] \mapsto [Y{:}X]$	$[Y^2{:}XY{:}X^2]$	b.p. / $[Y^2{:}X^2]$ (2:1)
Geometric role	the void	collapse	iso	conic	incompleteness
	$H^0 = 0$	trivial bundle	$\mathcal{O}(1)$ ample	Veronese	b.p. / cover

6 The Computational Pipeline

Every target above is *generated by the divisor* along two rails that are welded by the invertible sheaf. **Blue single arrows are constructions** ("build the next object from this one"); **amber double rungs are identifications** ("the same object in another language"). The top-left arrow "allow & force" is literally the valuation inequalities $\text{ord}_p(f) + n_p \geq 0$ of Target 1.



Top rail (analytic): $D \rightsquigarrow$ valuation inequalities $\rightsquigarrow L(D) = H^0 \to$ basis $\to \Phi_D$. *Bottom rail (local):* $D \to$ Cartier equations $f_\alpha \to g_{\alpha\beta} = f_\beta / f_\alpha \to \mathcal{O}(D) \to$ line bundle. The keystone $\mathcal{O}(D)$ identifies $\Gamma = H^0$ with the holomorphic sections of the bundle, and both rails terminate in the same Φ_D — once as $[s_0 : \cdots : s_r]$, once as the map of the complete system $|D|$.

7 Final Synthesis

A divisor is a way of interrogating a geometric situation, and the ledger is the transcript of the interrogation. Each abstract target was forced back to arithmetic: the divisor became valuation inequalities; $L(D)$ became an explicit space of rational functions; $\mathcal{O}(D)$ became local equations $f_\alpha = t_p^{n_p}$; the bundle became a single monomial transition $g_{01} = z^{-d}$; the sections became local representatives $a_\alpha = f_\alpha s$ that are honest to the trivialization; and the map became homogeneous polynomials whose common zeros (or absence thereof) decided base points and embeddings. Read across the five probes, the same machinery yields nothing ($\mathcal{O}(-1)$), a point ($\mathcal{O}$), a line ($\mathcal{O}(1)$), a conic ($\mathcal{O}(2)$), and — once we drop a section — a base point or a fold. To choose a divisor is to choose a question; to run its ledger is to receive the answer in the curve's own arithmetic.